Introduction to Numerical Analysis II, 5044, Spring 2025
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Homework set 1 due 02/07/2025 Name: Logan Reed

This worksheet is adapted from the homework sheets by Dr. Seibold.

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Please name your .m-files using your last name, the worksheet number, and problem number.

1. (*100 points*) Consider a Lotka-Volterra predator-prey model for a population of carps and pikes, whose numbers are given by $u(t)$ and $v(t)$, respectively. The growth/decay rates are given by

$$\begin{aligned} \frac{du}{dt} &= u - 4uv \\ \frac{dv}{dt} &= -v + 5uv \end{aligned} \tag{1}$$

(a) Show that the function $H(u, v) = uv \exp(-5u - 4v)$ is a *constant of motion*, i.e. if $(u(t), v(t))$ is a solution of (1), then $H(u(t), v(t))$ is constant in time.

(b) Using Matlabs `mesh` (or similar) command, plot the function H on the domain $(u, v) \in [0, 1]^2$.

(c) Using Matlab's `quiver` command, plot the velocity field given by the right hand side vector of (1), scaled to length 1 everywhere. On top of this plot, overlay isocontours of the function H, using Matlabs `contour` command.

(d) Starting with $u(0) = 0.2$ and $v(0) = 0.8$, approximate (1) using Eulers method for $t \in [0, 8]$. Use steps of size $\Delta t = 0.01$. Plot all 801 points obtained from this numerical solution into the figure created in (c). Explain why the resulting curve is not closed.

(e) Run the same computation with $\Delta t = 0.02$ and $\Delta t = 0.04$, as well as $\Delta t = 0.005$ and $\Delta t = 0.0025$. How does the observed approximation error behave? Explain your observations.

a)

Suppose u, v are solutions of (1).

Then

$$\frac{dH}{dt}(u,v) = \frac{d}{dt}\left[uv \exp(-5u-4v)\right]$$

$$= (uv)(-5u' - 4v')\exp(-5u-4v)$$

$$+ (u'v + v'u)\exp(-5u-4v)$$

$$= \Big[(uv)(-5u + \cancel{20uv} + 4v - \cancel{20uv})$$

$$+ (u - 4uv)v + (-v + 5uv)u\Big]\exp(-5u-4v)$$

$$= (-5u^2v + 4uv^2 + uv - 4uv^2 - uv + 5u^2v)\exp(-5u-4v)$$

$$= 0$$

Thus H is a constant of motion.

d) Since the global truncation error of Euler's method is $\mathcal{O}(\Delta t)$ $|y(t_{801}) - y_{801}|$ is bounded by a term proportional to Δt.

For the curve to be closed $y(t_{801}) = y_{801}$ which, while possible in general, is not guaranteed due to the aforementioned error bounds.

Also, due to the specific properties of the problem, the errors will result in a spiral outward.

e) The observed approx. behaviour behaves as expected, it decreases as $\Delta t \to 0$. This is expected because GTE is $\mathcal{O}(\Delta t)$ as previously mentioned, so the error bound decreases as $\Delta t \to 0$.